Student ID_____ Student Name_____

IS221: Web Applications Development

School of Information Technology, Engineering, Mathematics and Physics

Final Examination **Solutions**
Semester 1 2022

Face-To-Face and Online Mode

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 3 hours

INSTRUCTIONS

1. This examination has two sections:
   a. Section A: Coding Questions (40 marks)
   b. Section B: Short and Long Answer Questions (60 marks)
2. **All** questions are compulsory
3. Write **ALL** answers in the spaces provided within this paper.
4. This exam is **marked out of 100** and is worth **50%** of your **overall course marks**.
5. To **pass** this paper you require **at least 40% for this exam** which is equivalent to **40 marks out of 100**
6. This paper has 26 pages including this page.

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Section A: Coding 40 Marks

Question 1: ASP.Net

Web forms let you build interactive web pages that collect information from a user and process it on the web server. The Web page in figure 1 below shows an example of a typical web form. Use figure 1 web form and later information to answer the questions that follow.

Figure 1: Web Form

a. Write the ASP code for txtCusName Textbox. **(2 marks)**

```
<asp:TextBox ID="txtCusName" runat="server" "></asp:TextBox>
```

b. Write the ASP code for ddlLocation Dropdown List in asp.net with 3 List items (Ba, Lami, Lautoka). **(4 marks)**

Solution

```
<asp:DropDownList ID="ddlLocation" runat="server">
   <asp:ListItem>Ba</asp:ListItem>
   <asp:ListItem>Lami</asp:ListItem>
   <asp:ListItem>Lautoka</asp:ListItem>
</asp:DropDownList>
```

c. Bedroom option has a dropdown list (ddlBedrooms). The following are the options in ddlBedrooms: One, Two, Three, Four, Five or more. This list uses Bedrooms.xml as a data source. Write the XML code for Bedrooms.xml. **(5 marks)**

```
<?xml version="1.0" encoding="utf-8" ?>
<bedrooms>
    <bedroom name="One" />
    <bedroom name="Two" />
    <bedroom name="Three" />
    <bedroom name="Four" />
    <bedroom name="Five or more" />
</bedrooms>
```

d. Figure 2 below represent the RentHouse table in Microsoft Access. What control is used to connect the web form to the RentHouse table in Microsoft Access? **(1 mark)**

RentHouse

Field Name	Data Type
ID	AutoNumber
Cus_Name	Short Text
Cus_Phone	Short Text
Location	Short Text
Bedroom	Short Text
Pet_Friendly	Yes/No
Bills_Included	Yes/No

Figure 2: RentHouse Table

AccessDataSource

e. The customer phone number should only accept numeric values and have a total of seven digits.

i. What control should be used to validate txtCusPhone number? **(1 mark)**

RegularExpressionValidator

ii. Write down the validation expression to make sure the customer does not input any invalid phone number. **(1 mark)**

[0-9]{7} or ^ [0-9]{7}$

iii. What control should be used to force the customer to enter phone number before submitting the form? **(1 mark)**

RequiredFieldValidator

f. Upon clicking the "Request for Information" (btnRequestInfo) button, the customer information will be inserted into the database. Referring to figure 3 write the missing VB code that will handle btnRequestInfo click event. Part of the codes of the Event Handler is given below. **(7 marks)**

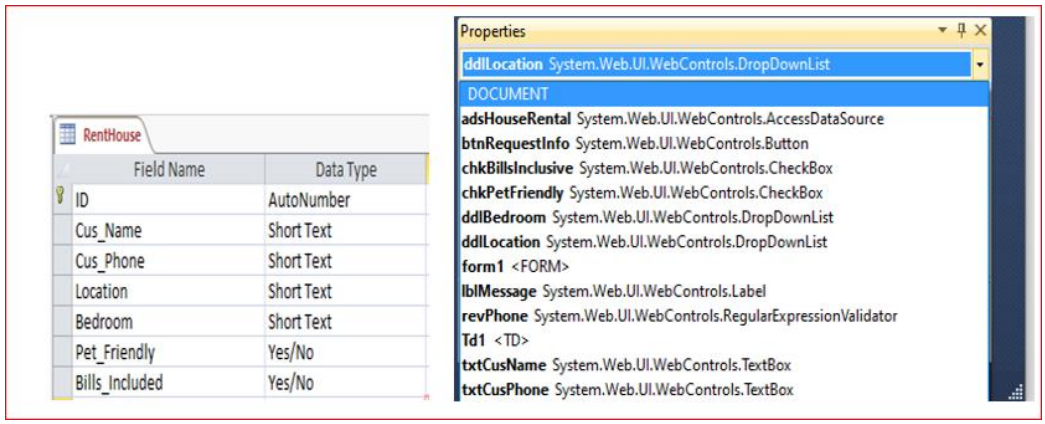

Figure 3

```
Protected Sub btnRequestInfo_Click(sender As Object, e As System.EventArgs)
Handles btnRequestInfo.Click

    ' write the missing VB codes below


  adsHouseRental.InsertParameters("Bedroom").DefaultValue = ddlBedroom.SelectedValue
  adsHouseRental.InsertParameters("Location").DefaultValue = ddlLocation.SelectedValue
  adsHouseRental.InsertParameters("Pet_Friendly").DefaultValue = chkPetFriendly.Checked
  adsHouseRental.InsertParameters("Bills_Included").DefaultValue = chkBillsInclusive.Checked
  adsHouseRental.InsertParameters("Cus_Phone").DefaultValue = txtCusPhone.Text
  adsHouseRental.InsertParameters("Cus_Name").DefaultValue = txtCusName.Text
  adsHouseRental.Insert()




    lblMessage.Text = "Thank you for choosing us. <br/> We will check the
    availability and get back to you soon."
  End Sub
```

g. Write the ASP code for Masoori Bungalows to have a printer friendly website by referencing the print.css file that will render successfully when a webpage is printed or viewed in print preview mode. **(2 marks)**

```
<link href="print.css" rel="stylesheet" type="text/css" media="print"/>
```

h. Write a CSS style to hide everything that is inside the nav element. **(1 mark)**

```
nav{
display: none;
}
```

i. Masoori Rento wants to make the website tablet friendly (780px and below). Write a media query that can be used to write CSS for tablets. **(2 marks)**

```
@media only screen and (max-width: 780px) {...}
```

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Question 2: Data Tables

The table below in Figure 4 shows Pet Medicine Product Price List for KOKOtree pet medicine products & Services Company.

a. Write the appropriate HTML codes that will markup the table in Figure 4 below to make meaningful semantic table.[Hint: consider all those tags we've covered: the table, the row, the data, the headers, the table head, table body, table foot, caption, the col group and the col, row tags]. **(7 marks)**

KoKotree Pet Medicine Product Price List

		Retail		Case		
Item #	**Product or Service**	**Price (ea.)**	**Retail Price (Case)**	**Case Discount**	**12+ Case Price**	**12+ Case Discount**
100050	Advance Pet Oral Care Toothbrush and Toothpaste	$9.55	$108.87	$5.73	$103.14	$11.46
100043	Basic Teeth Cleaning and Exam	$100.00	$1,140.00	$60.00	$1,080.00	$120.00
100013	Calm Cat Anxiety Relief Spray	$9.49	$108.19	$5.69	$102.49	$11.39
	Totals of these columns	$119.04	$1357.06	$71.42	$1285.63	$142.85

Figure 4

```
<table>
<caption>KoKotree Pet Medicine Product Price List</caption>
<colgroup>
<col class="item">
<col>
<col span= "2" class="retail">
<col span="3" class="case">
</colgroup>
<thead>
<tr>
<td></td>
<td></td>
<th colspan="2">Retail</th>
<th colspan="3">Case</th>
</tr>
<tr>
        <th scope="col">Item #</th>
        <th scope="col">Product or Service</th>
        <th scope="col">Price (ea.)</th>
        <th scope="col">Retail Price (Case)</th>
        <th scope="col">Case Discount</th>
        <th scope="col">12+ Case Price</th>
        <th scope="col">12+ Case Discount</th>
</tr>
</thead>
<tbody>
<tr>
        <td>100050    </td>
```

```
        <td>Advance Pet Oral Care Toothbrush and Toothpaste      </td>
        <td>$9.55 </td>
        <td>$108.87</td>
        <td>$5.73</td>
        <td>$103.14   </td>
        <td>$11.46</td>
    </tr>

    <tr>
        <td>100043    </td>
        <td>Basic Teeth Cleaning and Exam   </td>
        <td>$100.00 </td>
        <td>$1,140.00</td>
        <td>$60.00     </td>
        <td>$1,080.00</td>
        <td>$120.00</td>
    </tr>
    <tr>
        <td>100013    </td>
        <td>Calm Cat Anxiety Relief Spray     </td>
        <td>$9.49 </td>
        <td>$108.19   </td>
        <td>$5.69      </td>
        <td>$102.49   </td>
        <td>$11.39</td>
    </tr>
    <tbody>
    <tfoot>
         <tr>
         <td></td>
         <td>Totals of these columns</td>
         <td>$119.04</td>
         <td>$1357.06</td>
         <td>$71.42</td>
         <td>$1285.63</td>
         <td>$142.85</td>
        </tr>
        </tfoot>
    </table>
```

b. Write style rules that can be applied to the table in figure 4 by adding the following properties:

- All borders collapse.
- Table border: solid 1px #3F2B21
- Table cell bottom border: solid 1px #CCAE51
- Table cells padding: 0.5rem
- Caption: bold, left aligned, color #CCAE51, font size 2rem and padding of 1rem.

(6 marks)

```
table {
        border: 1px solid #3F2B21;
        border-collapse: collapse;
}
td, th {
        padding: 0.5rem;
        border-bottom: 1px solid #CCAE51;
}
caption {
  font-weight: bold;
  text-align: left;
  font-size: 2rem;
  padding: 1rem;
  color: #CCAE51
}
```

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Section C: Short/Long Answer Questions 40 Marks

Question 1

List three ways to control information overload when designing a web site. **(3 marks)**

Create manageable information segments, control page length, and use hypertext to connect facts, relationships, and concepts.

Question 2

Explain why you would include both graphic and text-based links on a web page. **(2 marks)**

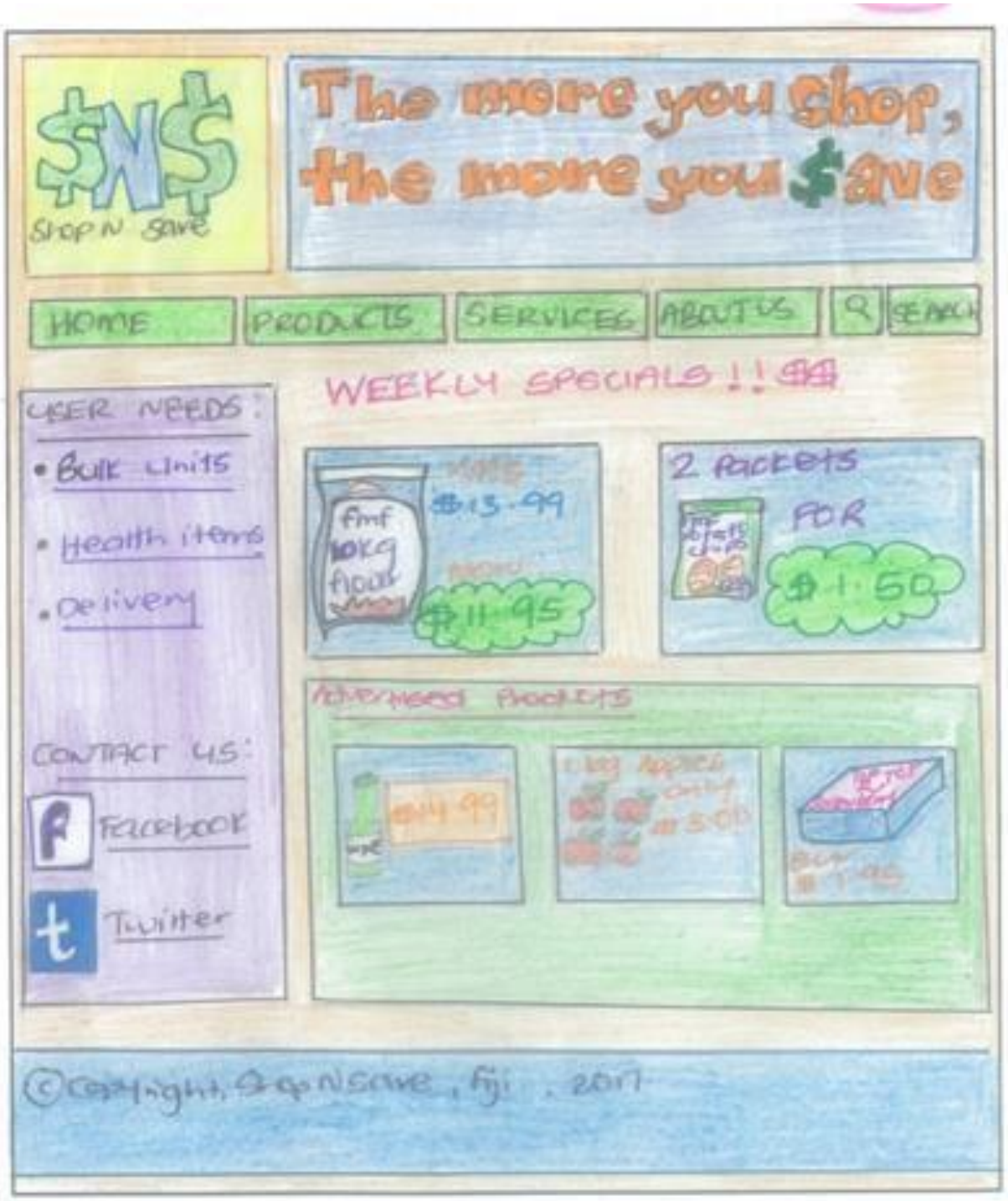
Text-based linking often is the most effective and accessible way to provide navigation on your site. It can work in both text-only and graphical browsers, and it does not depend on whether your images are displayed properly or not. Use linked graphics for navigation because they are more appealing.

Question 3

You have been hired to develop your local supermarket's website and are required to integrate the following site requirements:

- Logo and mission statement should be featured.
- Search box.
- Links to the other pages of the website should be available.
- Weekly specials should be a main feature content.
- Advertisements of related products, deals and news should also feature.
- Copyright statement should feature as well.
- Uniform look and feel for all the pages, the only difference should be the actual content.

a. With the aid of a diagram, illustrate how the home page template would look like **(5 marks)**


N
SHOP N SAVE
The more you shop, the more you $ave
HOME
PRODUCTS
SERVICES
ABOUT US
SEARCH
WEEKLY SPECIALS !!
USER NEEDS:
• Bulk units
• Health items
• Delivery
$13.99
$11.95
2 PACKETS
FOR
$1.50
CONTACT US:
Facebook
Twitter
© Copyright, Shop N Save, Fiji, 2017

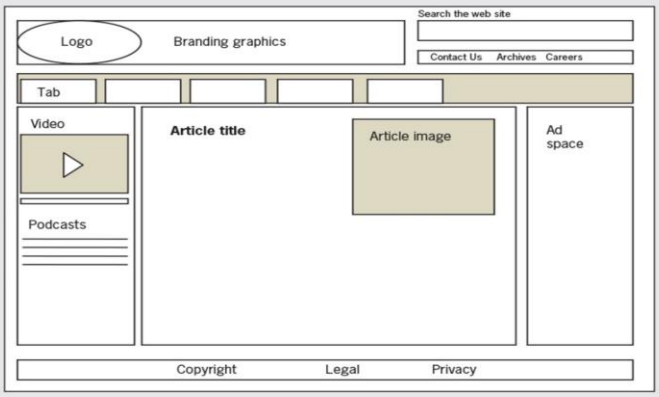


Figure 3-3: Sample wireframe for page layout
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b. With the aid of a diagram, illustrate how you would design a responsive version of the homepage for mobile devices that have screen sizes ranging from 320px to 768px.

(5 marks)

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Question 4

You have been asked to develop a website for a company that is proposing to sell flowers online. Using the web development lifecycle discuss how you would go about implementing this project. Your discussion should include the key stages of the cycle and examples of activities and artefacts involved in that stage. **(10 marks)**

Stage 1_____ (1/2 Mark)
Activities (1 Mark)
_____
_____
_____
Artefacts (1/2 Mark)
_____
_____

Stage 2_____ (1/2 Mark)
Activities (1 Mark)
_____
_____
_____
Artefacts (1/2 Mark)
_____
_____

Stage 3_____ (1/2 Mark)
Activities (1 Mark)
_____
_____
_____
Artefacts (1/2 Mark)
_____
_____

Stage 4_____ (1/2 Mark)
Activities (1 Mark)
_____
_____
_____
Artefacts (1/2 Mark)
_____
_____

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Stage 5_____ (1/2 Mark)
Activities (1 Mark)
_____
_____
_____
Artefacts (1/2 Mark)
_____
_____

Requirements specification

Activities

- Determine how the customer wants to use the site e.g. will there be credit card payments etc. Will there be a catalogue for customers to view and select the kava packs they want. How about delivery, will you customers need to provide their delivery address etc.

- Type of website in this case will be an e-commerce website therefore think of common elements that an e-commerce website should have.
- Since it's going to be an e-commerce website you will need to think of an appropriate domain name e.g. renting online a .com domain would be most appropriate.
- Confirm requirements and specifications with client. Artefacts

- Registered domain name (www.uniroomsforrent.com.fj)
- Requirements specifications document which will consist of: ○ Technical specifications
  e.g. online searchable catalogue, form to enter customer's details etc and a database to store catalogue details and if need be customers' details.
  - *Design requirements* e.g. colour scheme or a theme to be used for the website that is in line with the Uni rooms for rent concept
  - *Page layout sketches* e.g. a sketch of what the site may look like ○ *Audience definition* e.g. who your potential customers are, will they be just local customers or international customers

Information design and taxonomy creation

` Activities

- Sorting out the web pages the customer wants for instance what content to have on what page and how to structure content and hyperlinks etc. Think in terms of the layouts discussed in readings, customer might want to have a featured product/rental specials or something

- Sketch of what the pages and the website may look like and navigation between pages etc.

Artefacts

- A sketch of the various web pages and how they should be interlinked based on their content

Graphic design and page template creation

Activities

- Creating web graphics, and page templates e.g. logo for Uni rentals should be representative of their product and what the company wants to portray etc.

- Navigation buttons/scheme etc.
- Wireframe sketches of web site/web pages Artifacts5

- Web graphics, web page templates

Construction and Web development/Quality Assurance and User Testing

Activities

- Coding of the required elements and content

- Construction of databases
- Testing of coded elements to ensure they are functioning correctly e.g. testing the room booking functionality etc.

Artefacts

- A database driven functioning website with forms, catalogues and components that are tested and functional.

Publishing and promotion

Activities

- Determining whether to host the website onsite (your own servers etc) or use a web hosting company.

- Web site is uploaded to web server and is now available and live

- Use Facebook for business to promote website or twitter etc Artefacts

- Fully functional website

Maintenance

Activities

- Updating the website e.g. adding new products, on sale items etc.

Artefact

- Well maintained site that is available, up to date and fully functional

Question 5

Suppose you are part of the web designing team for the university's website. How would you go about designing and developing USP's website so that it is easy to maintain and update; yet each school can customize the design to reflect their school colors and requirements. **(5 marks)**

- *Determine common content to be put on Master page*
- *Create Master page*
- *Use external stylesheets to style the Master page*
- *Link external stylesheet to Masterpage*
- *Use Content Place Holders in Master page for any sections that should be editable.*
- *Create Nested Master pages for the various schools and stylesheets.*
- Determine common elements for all pages
- Create master page integrating common elements
- Use content place holders for sections that will be editable
- Create stylesheet for master pages using CSS
- Link stylesheet to Masterpage
- Generate web pages using master page

Question 6

ASP.NET applications processing is depicted in figure3 . Using the labels 1 – 5, describe this process. **(5 marks)**

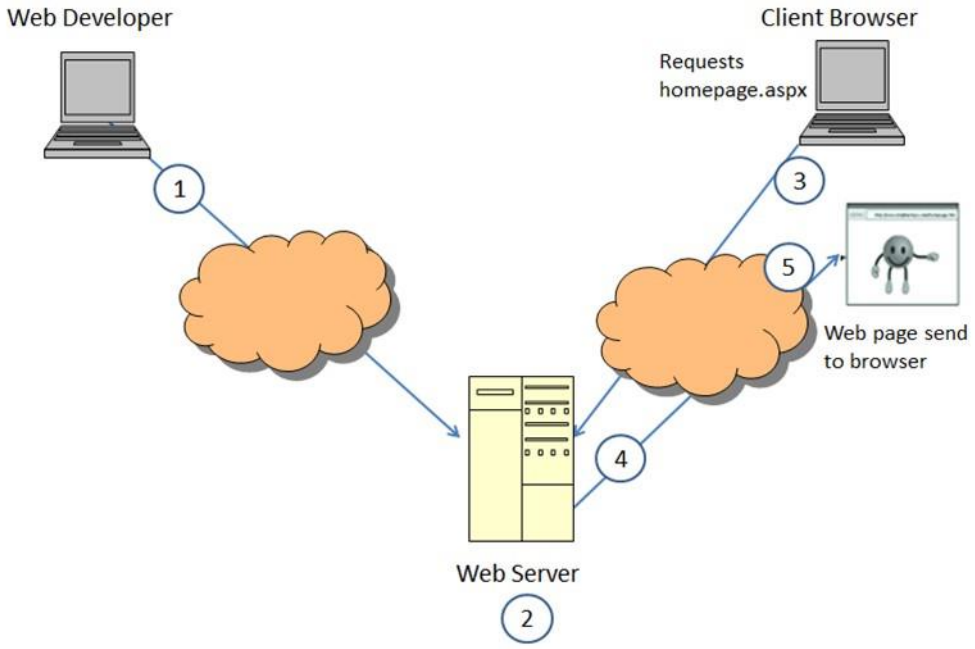

Figure 5: ASP.Net applications processing

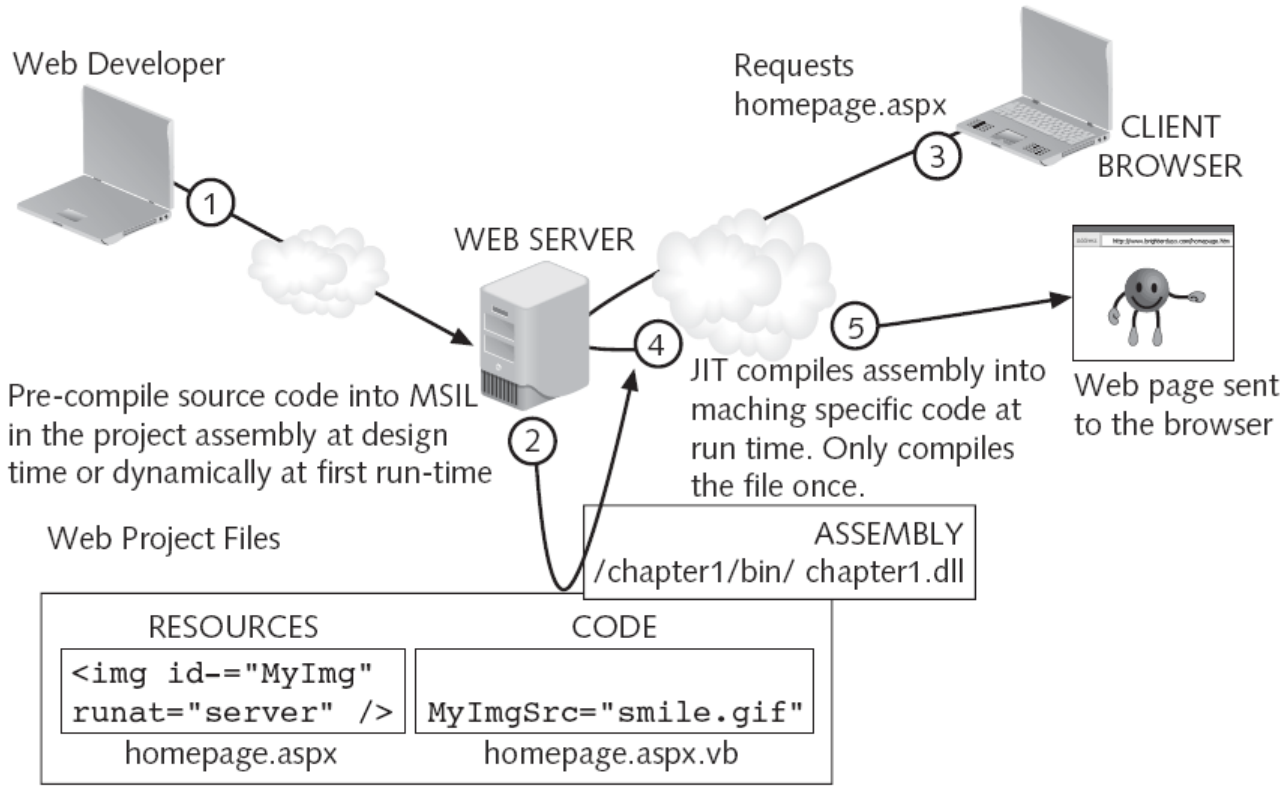

Figure 1-21 Processing an ASP.NET application

- Web Forms are web pages identified with the file extension .aspx
- Assembly contains language- and computer-independent representation of code called Microsoft Intermediate Language (MSIL)
- ASP.NET engine dynamically compiles the assembly and translates into computer-specific instructions
- HTML output sent back to the browser

Question 7

Effective navigation should provide information that helps the user to answer the following questions:

- Where am I?
- Where can I go?
- How do I get there?
- How do I get back to where I started?

Compare and contrast how Facebook and YouTube implemented their navigation schemes and how they provided answers to the stated questions. For each question determine the required information to be provided and how each of the sites provided the answers.

(8 marks)

Where am I?

Required information
The current page and the type of content they are viewing

_____
_____
_____

Facebook

Bread crumps

_____
_____
_____

YouTube

Bread crumps / site map

_____
_____
_____

Where can I go?

Required information
Where they are in relation to the rest of the Web site

_____
_____
_____

Facebook

_____
_____
_____

YouTube

_____
_____
_____

How do I get there?

Required information
Consistent, easy-to-understand links

_____
_____
_____

Facebook _
site map /contextual links text / graphics based navigation / not more than 3 clicks
_____
_____
_____

Vodafone
site map _____
_____
_____

How do I get back to where I started?

Required information
Alternatives to the browser's Back button that lets users return to their starting point

_____
_____
_____

Facebook ___ page turner's
_____
_____
_____

YouTube _____- linked navigation
_____
_____
_____

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Question 8

Using good coding practices ensures the greatest standards-compliance, presentation, and usefulness of the content. List with examples five rules that conforms to good coding practices. **(5marks)**

ANSWER:

1. All tag names must be in lower case -

   <html> NOT <HTML>

2. All tags must have a corresponding matching closing tag (container tags)

   <body> </body>

3. Empty elements must have an end tag or include the closing tag

 </br> OR

4. All attribute values must be enclosed in quotes (either double or single – the notes use double)

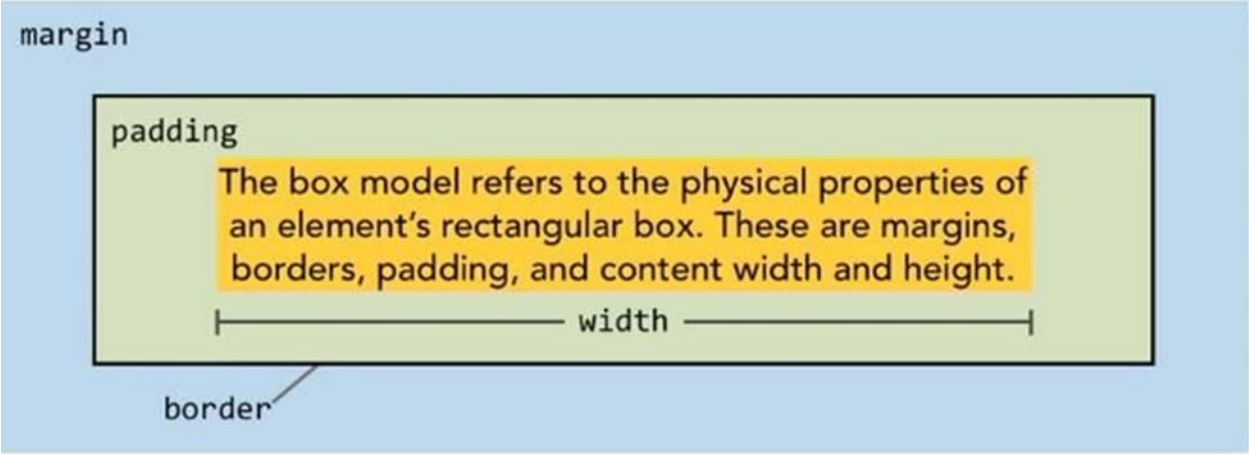
   size="10"

5. Nested tags must be terminated in the reverse order to the way they are written.

   a. … is a correct construction,
   b. … is incorrect.

Question 9

Using a diagram explain CSS box model. **(5 marks)**



Question 10

Discuss how you would design for both Accessibility and Legibility **(2 Marks)**

Accessibility

Information and user interface should be perceivable to the users

User interface components should be operable by users

Information about the user interface its operations must be understandable by users Content should be robust enough to be interpreted by wide variety of agents.

Legibility

Choice of fonts should be easy to read and universally available for users' browsers.

Choice of color should complement content

Use graphics sparingly for emphasis

Use of active whitespace for structure and visual enhancement of text

Question 11

Moodle content management learning system is very compelling to use, attractive, easy to navigate and a very popular content management website. In designing the Moodle, the designers took into consideration the following pointers:

- Crafting the look and feel
- Design portability
- Easy access to your information
- Planning for Clear presentation of information
- Design for low bandwidth

For each consideration discuss how Moodle content management website implemented them. **(5Marks)**

Crafting the look and feel (1 mark)

Grid structure, uniform use of blue and white color scheme and clearly labelled sections etc. afforded a very professional look that is intuitively easy to use.

Design portability (1 mark)

Coding to standards and the use of different stylesheets etc. enable Facebook's site to be accessible on any platform. They also have a mobile version of the main site.

Easy access to your information (1 mark)

Facebook used a great information design that is based on what the users are expected to be able to do and as such structured the main links accordingly hence maintaining the 3-click rule of accessing information.

Planning for Clear presentation of information (1 mark)

Use of the grid structure once again and active use of white space that partitions the various sections of FB with clear headings and icons that are representative of the various activities e.g. mail icon for messaging, etc.

Design for low bandwidth (1 mark)

Minimalistic use of graphics and strategic reusing of universally understood icons.

(5 Marks)

THE END